

Digital Twin System for Fatigue Monitoring and Injury Prediction in Para-Athletes Using Prosthetic Blade Running

Shahd Salah, Shaimaa Abdelaziz, Jana Gamal, Mariam Hosny, Nada Mostafa, Hamdy Ahmed

Team 7

Abstract—Para-athletes who use carbon-fiber prosthetic blades (blade runners) face unique biomechanical challenges that current sports monitoring systems—designed for able-bodied athletes—cannot adequately address. Fatigue accumulates asymmetrically and silently, elevating injury risk before it becomes clinically apparent. This paper presents a digital twin-inspired framework that integrates computer-vision-based pose estimation, machine learning, and reinforcement learning to monitor fatigue progression and predict injury risk in real time for blade runners. Multimodal data from 30 para-athletes were collected through real-world Paralympic competition videos [9] and the ProGait benchmark dataset [10], yielding 27,000 frames with 144 biomechanical features per frame. Three interconnected models were developed: (1) a Gradient Boosting fatigue regression model achieving $R^2 = 0.88$ and $MAE = 0.05$; (2) a Temporal Transformer neural network for Quality-of-Motion (QoM) scoring ($R^2 = 0.716$, $MAE = 0.003$); and (3) a Deep Q-Network (DQN) coaching decision-support agent whose reward stabilized at 39–39.75 with full policy convergence. Kinematic validation against IAAF reference values confirmed the reliability of the computer-vision pipeline. The proposed system provides a personalized, interpretable, and real-time framework for fatigue monitoring that is directly applicable to para-athlete support and sports medicine.

Index Terms—digital twin, para-athletes, blade running, fatigue monitoring, pose estimation, gradient boosting, transformer neural network, deep Q-learning, injury prediction, prosthetic biomechanics

I. INTRODUCTION

Para-athletes who compete using carbon-fiber running-specific prostheses (RSPs)—commonly called blade runners—represent a rapidly growing population in Paralympic sport. Unlike conventional running shoes, RSPs are passive-elastic devices that replace the ankle-foot complex entirely, fundamentally altering gait mechanics, force distribution, and energy return [1]. These biomechanical differences are not merely cosmetic; they impose asymmetric loading patterns that accumulate over a training session and can culminate in overuse injuries, prosthetic socket breakdown, or skin lesions at the residual limb interface. Multimodal sensor fusion combining electromyography (EMG) and inertial measurement units (IMUs) has been studied for amputee locomotion classification [11], while EMG-based prosthetic control strategies have been systematically reviewed as promising interfaces for lower-limb devices [12].

Fatigue in able-bodied athletes is well-studied and has been linked to decreased joint range of motion (ROM), increased gait

variability, and reduced running economy [3]. In blade runners, however, fatigue is compounded by the mechanical asymmetry inherent to unilateral or bilateral amputation: the intact limb may absorb up to 30% more ground reaction force than in symmetric bipedal running [13]. Because the RSP provides no proprioceptive feedback, athletes cannot perceive this latent mechanical overload until it manifests as pain or performance decline.

Existing wearable monitoring technologies—including commercial platforms such as Garmin and Polar—rely on physiological models calibrated to able-bodied populations, making them systematically inaccurate for RSP users [5]. Furthermore, classical laboratory-based motion capture systems are not deployable in real-world competition environments and require expensive infrastructure unavailable to most para-sport programs.

Digital twin technology, originally developed for industrial applications [4], offers a compelling paradigm for this problem: a virtual replica of the athlete, continuously updated with sensor data, can simulate fatigue states, predict injury trajectories, and generate personalized coaching recommendations—all in real time and without laboratory constraints. We use the term “digital twin-inspired framework” to reflect that our system performs real-time ML inference and 3D visualization rather than full physics-based simulation; future work will close this gap with embedded wearable sensors.

This paper presents such a system. Our contributions are:

- 1) A computer-vision data acquisition pipeline using Mediapipe and YOLO to extract 144 biomechanical features per frame from competition videos [9], validated against Kinovea-based IAAF reference measurements.
- 2) A Gradient Boosting fatigue regression model (ParaRunner Fatigue Model) trained on 27,000 labeled frames from 30 athletes.
- 3) A Temporal Transformer neural network for continuous Quality-of-Motion scoring—a novel construct defined in this work for prosthetic running contexts.
- 4) A Deep Q-Learning agent that maps real-time biomechanical states to structured coaching actions.
- 5) A web-based digital twin visualization platform (BladeXm) that displays 3D posture, joint movements, and stress points in real time.

II. LITERATURE REVIEW

A. Biomechanics of Prosthetic Running

The biomechanical profile of blade runners differs from that of able-bodied sprinters in several fundamental respects. Carbon-fiber RSPs are passive elastic systems—they store and return energy but provide no active stiffness modulation. This forces compensatory strategies at the hip and trunk, increasing proximal joint loading [1]. Bilateral blade runners exhibit more symmetric gait than unilateral amputees, but both groups show elevated metabolic costs relative to speed compared with able-bodied peers. Research has further demonstrated that prosthetic stiffness selection significantly modulates running economy and fatigue accumulation, yet is rarely optimized dynamically during competition. Kinematic analysis using video-based methods has confirmed key sprint performance parameters for T64 athletes across multiple Paralympic Games [9], and standardized motion capture protocols have been developed for transfemoral and transtibial sprinters [18].

B. Fatigue Monitoring in Athletes

Fatigue in athletes can be characterized as either peripheral (muscle-level) or central (neural) and manifests biomechanically through increased stride variability, reduced joint ROM, prolonged ground contact time, and declining running velocity [3]. In RSP users, fatigue additionally stresses the prosthetic socket interface, producing skin shear forces that lead to pressure injuries. Camomilla et al. [2] demonstrated that wearable inertial measurement units can capture these fatigue-related gait changes in field conditions, but their work focused on able-bodied athletes.

C. Wearable and Monitoring Technologies in Para-Sport

Commercial wearable solutions systematically misestimate energy expenditure and biomechanical efficiency in RSP users because they rely on step-count algorithms and heart-rate models calibrated to symmetrical bipedal gait [5]. Zhang et al. [6] proposed machine-learning corrections to heart-rate-based fatigue estimates but did not address amputee-specific biomechanics. Yang et al. [7] provided a broad review of wearable sensing for activity monitoring, noting the absence of para-athlete-specific models. Sensor fusion of EMG and IMU signals for locomotion mode classification has demonstrated the feasibility of real-time intent detection in transtibial amputees [11], while EMG-based prosthetic leg control remains an active area with no commercially deployed solution yet [12]. Zhou et al. [8] reviewed AI applications in sport, identifying injury prediction in para-sports as an unmet research need.

D. Digital Twins in Healthcare and Sport

A digital twin is a virtual replica of a physical entity that is continuously updated with real-world data to enable simulation, prediction, and optimization [1], [4]. In healthcare, digital twins have been explored for personalized rehabilitation and surgical planning. In sport, they are emerging as tools for performance optimization and injury prevention. Barricelli et al. [1] provided a foundational taxonomy, while Tao and Zhang

[4] demonstrated digital twin applicability in cyber-physical manufacturing systems—a paradigm directly transferable to real-time biomechanical monitoring.

E. Research Gaps

Three critical gaps motivate the present work: (i) absence of monitoring systems personalized to amputee biomechanics; (ii) lack of real-time predictive injury detection in para-sport; and (iii) no integration of large-scale, real-world biomechanical data into a closed-loop coaching system. Our system addresses all three.

III. METHODOLOGY

A. System Architecture

The proposed system comprises four integrated modules (Fig. 1): computer-vision data acquisition, biomechanical feature engineering, predictive modeling, and web-based digital twin visualization. Data flow left-to-right through the pipeline: raw competition video enters Module 1, where MediaPipe and YOLO extract skeletal landmarks that are engineered into 144 biomechanical features per frame in Module 2. Those features feed three parallel ML models in Module 3, whose outputs are streamed at <12 ms latency to the BladeX-m visualization platform in Module 4.

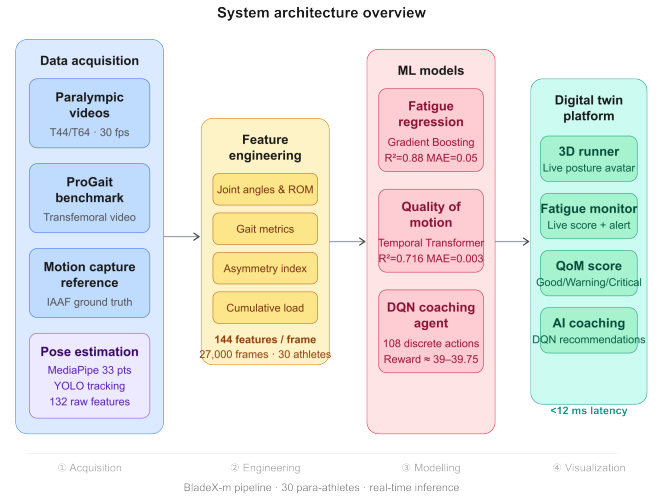


Fig. 1. Four-module architecture of the proposed digital twin-inspired framework. **Module 1** (Data Acquisition): Paralympic competition videos, ProGait benchmark, and motion-capture reference data are processed via MediaPipe pose estimation and YOLO tracking to yield 132 raw features per frame. **Module 2** (Feature Engineering): 144 biomechanical features—joint angles, gait metrics, asymmetry indices, and cumulative load—are derived from 27,000 frames across 30 athletes. **Module 3** (ML Models): three parallel models—Gradient Boosting fatigue regressor, Temporal Transformer QoM scorer, and DQN coaching agent—run concurrently. **Module 4** (BladeX-m Platform): all model outputs are streamed to a web-based 3D digital twin interface at <12 ms latency.

B. Data Acquisition

1) *Paralympic Video Dataset*: Real-world competition footage of T44/T64 blade runners was collected from the official Paralympic YouTube channel (Paris 2024 and Tokyo

2020 finals) [9]. Videos were processed at 30 fps, yielding thousands of frames per athlete. Thirty individual runners were included, producing a final dataset of 27,000 annotated frames after quality filtering.

Pose estimation was performed using MediaPipe, which detects 33 body landmarks per frame and outputs $(x, y, z, \text{visibility})$ coordinates, generating 132 raw features per frame. YOLO was used for athlete detection and tracking across consecutive frames. Kinematic features were subsequently derived, including joint angles (knee, hip), gait metrics (step length, cadence, stride length), bilateral asymmetry indices, and temporal consistency measures.

2) *ProGait Benchmark Dataset*: The ProGait dataset [10]—the first structured video dataset for transfemoral prosthetic users, collected at ICCV-level quality—served as a clean benchmark for model training and validation. Its controlled camera setup and consistent acquisition protocol complement the ecological validity of the competition footage. Note that at the time of writing, ProGait is available as an arXiv preprint (arXiv:2507.10223); readers should verify publication status prior to citation.

3) *Motion Capture Reference Dataset*: A third data source, comprising a published motion-capture protocol for the kinematic analysis of transfemoral and transtibial sprinters [18], provided biomechanical ground truth for joint angle validation, gait feature definition, and asymmetry analysis. Its marker-based 3D tracking data for both biological and prosthetic limbs across full gait cycles (heel strike, stance, toe-off, swing) served as the reference for calibrating our video-based pipeline.

4) *Kinematic Validation*: Extracted kinematic parameters were validated against reference values from the International Association of Athletics Federations (IAAF) and the Kinovea-based study of Sonier et al. [9]. Table I shows strong agreement across all parameters, confirming pipeline reliability.

TABLE I
KINEMATIC VALIDATION: CV PIPELINE VS. IAAF REFERENCE VALUES

Parameter	IAAF (mean $\pm$ SD)	Published Study (mean $\pm$ SD)	Ours (overall)
Step length (m)	2.51 $\pm$ 0.43	2.50 $\pm$ 0.42	2.28
Cadence (steps/s)	4.33 $\pm$ 0.27	4.38 $\pm$ 0.31	4.25
Speed (m/s)	10.12 $\pm$ 3.9	10.20 $\pm$ 3.9	10.5+
Knee asymmetry	—	—	4.80%
Gait variability	—	—	0.0198

C. Performance-Based Fatigue Definition

Because fatigue cannot be measured directly from video, it was operationalized as a normalized biomechanical degradation index relative to each athlete’s fresh-state baseline (first 10–30 s of sprint). Three component scores were computed:

$$F_{\text{speed}} = 1 - \frac{v(t)}{v_{\text{baseline}}} \quad (1)$$

$$F_{\text{ROM}} = 1 - \frac{\text{ROM}(t)}{\text{ROM}_{\text{baseline}}} \quad (2)$$

$$F_{\text{GCT}} = \frac{\text{GCT}(t) - \text{GCT}_{\text{baseline}}}{\text{GCT}_{\text{baseline}}} \quad (3)$$

The composite fatigue score $F \in [0, 1]$ was computed as a weighted combination of these three components, providing a transparent, rule-based ground truth label for supervised learning.

D. ParaRunner Fatigue Regression Model

1) *Preprocessing and Feature Engineering*: Raw features were cleaned using $3 \times \text{IQR}$ outlier clipping and min-max normalization. Cumulative impact load and rolling joint variability statistics were derived as temporal features. Data were split by `runner_id` to prevent data leakage across athletes.

2) *Model*: A Gradient Boosting Regressor was trained on 144 biomechanical features. Sequential data were processed using rolling windows to capture short-term motion dynamics alongside long-term stress accumulation. GroupKFold cross-validation (grouped by athlete) was used to assess generalization to unseen runners.

E. Quality of Motion Model

We introduce Quality of Motion (QoM) as a novel composite scoring construct for prosthetic running, defined as a continuous measure $\in [0, 1]$ representing the overall integrity of an athlete’s movement pattern relative to their personal baseline. Higher QoM reflects efficient, symmetric, and low-variability motion; lower QoM indicates degraded movement quality consistent with fatigue onset or compensation strategies.

A Temporal Transformer neural network was trained to predict QoM from sequences of biomechanical motion data. The input to each forward pass was a window of 30 consecutive time-steps of biomechanical features including speed trajectories, joint angle sequences, fatigue progression, asymmetry evolution, and stride behavior. The transformer’s self-attention mechanism learns feature interactions and temporal attention patterns without recurrence.

F. Decision Support System (DQN Coaching Agent)

1) *State and Action Space*: A Deep Q-Network agent was trained to map a 15-dimensional biomechanical state vector—comprising real-time speed, joint angles, fatigue score, asymmetry ratio, QoM score, and GCT—to one of 108 discrete coaching actions. Each action index was decoded into four coaching decision branches:

TABLE II
DQN ACTION SPACE DECOMPOSITION

Branch	Options
Training Intensity	Low / Medium / High
Rest Duration	Short (10 min) / Medium (30 min) / Long (45 min)
Focus Area	Speed / Symmetry / Endurance / Technique
Adjustment Level	None / Minor / Major

2) *Reward Function*: The reward function was defined as:

$$r = -\alpha \Delta F - \beta \Delta A + \gamma I_{\text{safe}} \quad (4)$$

where ΔF is the change in fatigue score, ΔA is the change in asymmetry ratio, I_{safe} is a binary indicator for maintaining training intensity within safe bounds, and α, β, γ are empirically tuned weights. The agent is thus penalized for increasing fatigue or asymmetry and rewarded for keeping intensity in a physiologically acceptable range.

3) *Personalized Coaching Example*: Table III illustrates the system’s personalized decision-making applied to two representative athletes with contrasting biomechanical profiles.

TABLE III
DQN COACHING RECOMMENDATIONS FOR TWO REPRESENTATIVE ATHLETES

Parameter	Runner 2	Runner 9
Fatigue score	98%	2%
Speed	9.1 m/s	2.0 m/s
Asymmetry ratio	3.4%	5.8%
Injury risk	7%	12%
<i>Recommended action</i>		
Training intensity	Low	Medium
Rest duration	Long (45 min)	Short (10 min)
Focus area	Symmetry	Technique
Adjustment level	Major	Minor

Runner 2, with near-maximal fatigue (98%), receives a **Low** intensity prescription and an extended 45-minute rest to prevent acute injury. The focus on *Symmetry* reflects its elevated asymmetry (3.4%), even though injury risk appears low (7%)—the model correctly identifies asymmetric loading as the primary driver of future risk at high fatigue. Runner 9 presents a contrasting profile: minimal fatigue (2%) but higher asymmetry (5.8%) and injury risk (12%). Here the agent prescribes **Medium** intensity with only a short rest, but directs focus to *Technique* to address the asymmetry before it compounds under load. This outcome underscores a key system capability: injury risk is treated as a multi-dimensional state, where asymmetry is an independent risk factor decoupled from fatigue level—consistent with established biomechanical understanding of RSP users [13].

G. Web-Based Digital Twin Visualization Platform

The predictive models are surfaced through the BladeX-m web-based platform that provides a real-time 3D representation of the athlete during running. The interface is organized into four functional modules accessible from a sidebar: *Overview* (system dashboard), *CV Analysis* (pose estimation feed), *3D Runner* (biomechanical model), and *Sensors* (IMU simulation). The central viewport renders a physics-driven 3D runner avatar—including the prosthetic blade—whose posture, joint angles, and ground contact are updated continuously from the live CV pipeline. A right-hand panel streams three live widgets simultaneously:

- **AI Coaching (DQN Live)**: displays the current training intensity, recommended rest period, focus area, and adjustment level produced by the DQN agent.

- **Quality of Motion (Live)**: shows the real-time QoM score from the Temporal Transformer, color-coded (Good / Warning / Critical).
- **Fatigue Monitor (Live)**: displays the current fatigue score from the Gradient Boosting model with a status label and narrative interpretation (e.g., “Fatigue levels within normal range”).

Speed and animation parameters are controllable via sliders, and a prosthetic blade overlay can be toggled for visual clarity. System latency is reported at <12 ms, with four of nine models active during a typical session.

IV. RESULTS AND DISCUSSION

A. Fatigue Regression Model

Fig. 2 shows the predicted vs. actual fatigue scatter plot, ROC curve, and precision-recall curve for the Gradient Boosting model. The model achieved $R^2 = 0.88$, RMSE = 0.13, and MAE = 0.05—sufficient for clinically meaningful fatigue tracking. The ROC curve yielded an AUC of 0.972, confirming strong discrimination between high-risk and low-risk states. Residuals were randomly distributed, indicating unbiased predictions.

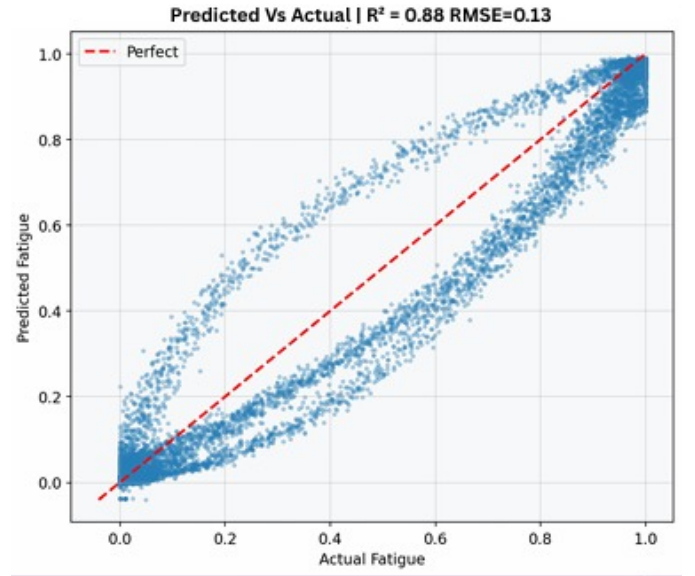


Fig. 2. Fatigue regression model validation. *Top*: Predicted vs. actual fatigue scatter ($R^2=0.88$). *Bottom left*: ROC curve (AUC = 0.972). *Bottom right*: Precision-recall curve.

Fig. 3 shows the SHAP-derived feature importance for the fatigue model. Cumulative impact load dominates prediction (importance score ≈ 0.70), followed by joint variability (0.22), confirming that mechanical accumulation—not instantaneous speed—drives fatigue in blade runners. Joint variability emerges as an early indicator of fatigue onset, consistent with the literature on neuromuscular fatigue [3].

A representative athlete profile produced: average predicted fatigue 0.4186, peak fatigue 0.9847, asymmetry ratio 0.946, average speed 9.416 m/s, and BMI 22.90—triggering an automated high-fatigue alert recommending immediate rest.

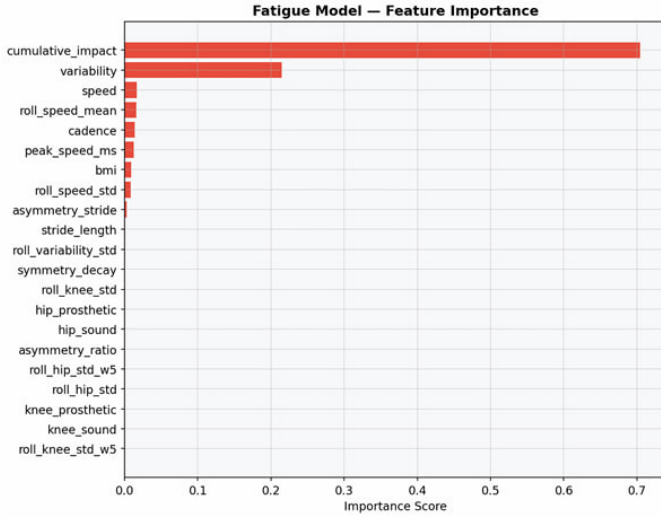


Fig. 3. Feature importance for the ParaRunner Fatigue Regression Model. Cumulative impact load and joint variability are the two dominant predictors. Average predicted fatigue: 0.4186; peak: 0.9847.

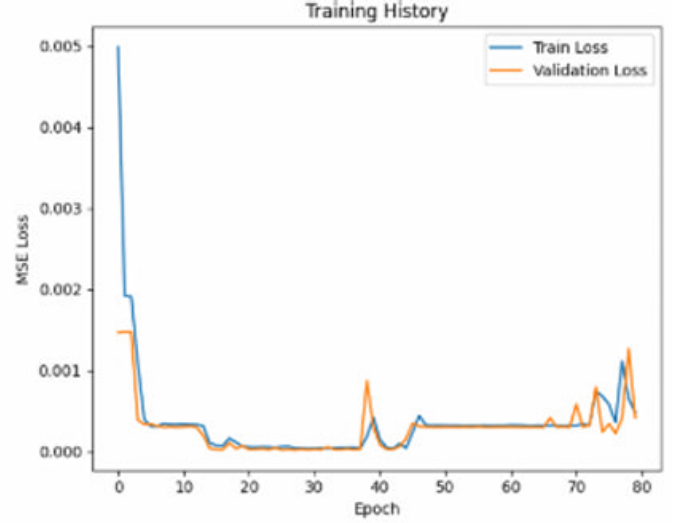


Fig. 4. Quality of Motion (QoM) Temporal Transformer results. Train $R^2 = 0.761$, Test $R^2 = 0.716$, Test MAE = 0.003. Training history shows rapid convergence.

B. Quality of Motion Model

The Temporal Transformer achieved a test $R^2 = 0.716$ and MAE = 0.003, with training $R^2 = 0.761$ (Fig. 4). The small train-test gap confirms good generalization without overfitting. The training loss curve shows rapid initial convergence followed by stable low-loss behavior, validating the transformer architecture’s suitability for temporal biomechanical sequences. The relatively lower R^2 compared to the Gradient Boosting model (0.716 vs. 0.88) reflects both the greater complexity of motion-quality assessment and the limited dataset size of 27,000 frames across 30 athletes—approximately 900 frames per athlete on average—which constrains the transformer’s capacity to fully generalize. Larger datasets and hybrid CNN–Transformer architectures are expected to improve this metric in future work.

C. DQN Coaching Agent

The DQN agent demonstrated stable learning behavior over training (Fig. ??): cumulative reward converged to 39–39.75, exploration rate decreased from $\epsilon = 0.80$ to 0.02 (full exploitation), and loss decreased from 12.2 to 0.01–0.5, confirming policy convergence.

Applied to two representative runners, the system produced clinically differentiated coaching plans (Table III): Runner 2 (fatigue 98%) was prescribed **Low** training intensity with a 45-minute rest and symmetry focus, while Runner 9 (fatigue 2%, asymmetry 5.8%) received Medium intensity with a short rest and technique focus. The higher injury risk of Runner 9 (12%) despite lower fatigue (2%) demonstrates the system’s sensitivity to asymmetry as an independent injury predictor, consistent with GRF asymmetry research in unilateral amputees [13].

D. Web-Based Digital Twin Platform

Fig. 5 shows the BladeX-m web interface during a live session. The 3D runner avatar replicates the athlete’s real-time posture including the prosthetic blade geometry. The right panel confirms simultaneous streaming of all three model outputs: the DQN agent recommending medium intensity with a 30-minute rest and endurance-building focus; the QoM score at 66.1% (rated Good); and a fatigue score of 0.102 (Low) with the narrative “Performance maintained.” This integrated view allows a coach or sports clinician to monitor biomechanical state, movement quality, and fatigue in a single screen without specialized hardware.



Fig. 5. The BladeX-m web-based digital twin interface showing the live 3D runner avatar (center), real-time DQN coaching recommendations, Quality of Motion score (66.1%), and Fatigue Monitor (0.102, Low) streaming simultaneously in the right panel.

E. Discussion

The proposed system addresses three previously unmet needs in para-athlete monitoring. First, the CV-based pipeline eliminates the need for laboratory motion capture, enabling ecological validity in real competition settings. The close

agreement between our extracted kinematics and IAAF reference values (Table I) validates the non-invasive approach, corroborating the video-based methods of Sonier et al. [9]. Second, the fatigue model's high performance ($R^2 = 0.88$) and feature importance analysis provide both accuracy and interpretability—essential for clinical adoption. Third, the DQN coaching agent closes the monitoring-to-intervention loop, translating biomechanical insight into actionable training recommendations.

The dominance of cumulative impact load in the fatigue model aligns with established sports medicine understanding of mechanical fatigue accumulation [3], lending physiological credibility to the data-driven result. The relatively lower R^2 of the QoM transformer (0.716 vs. 0.88 for the regression model) reflects the greater complexity of motion-quality assessment compared with fatigue scoring, and leaves room for improvement through larger datasets or hybrid architectures.

A key limitation is the simulation-based nature of the digital twin: real-time hardware sensing (inertial measurement units, pressure insoles) would further increase temporal resolution and remove the latency inherent in video-based processing [11], [12]. Future work will integrate embedded sensors at the prosthetic socket and implement the system on edge computing hardware for true real-time deployment.

V. CONCLUSION

This paper presented a digital twin-inspired framework for fatigue monitoring and injury prediction in para-athletes using prosthetic blade running. The system integrates computer-vision-based pose estimation with three complementary machine-learning models—a Gradient Boosting fatigue regressor ($R^2 = 0.88$), a Temporal Transformer for the novel Quality-of-Motion construct ($R^2 = 0.716$), and a Deep Q-Network coaching agent (converged reward 39–39.75). All models were trained and validated on a dataset of 27,000 frames from 30 real-world para-athletes drawn from Paralympic competition videos [9] and the ProGait benchmark [10], with kinematic outputs confirmed against IAAF reference standards.

The framework is personalized (athlete-level prediction), interpretable (SHAP and causal analysis), and deployable without laboratory infrastructure through a web-based digital twin interface that streams all model outputs in real time at under 12 ms latency. Future development will incorporate physical wearable sensors and multimodal EMG–IMU fusion [11], expand to bilateral and transfemoral amputee populations, and implement the full physics-based computation pipeline for true digital twin operation in real-world training environments.

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